Effect of Captopril on Progression to Clinical Proteinuria in Patients With Insulin-Dependent Diabetes Mellitus and Microalbuminuria

Giancarlo Viberti, FRCP; Carl Erik Mogensen, MD; Leif C. Groop, MD; John F. Pauls, MS;
for the European Microalbuminuria Captopril Study Group

Objectives.—To study the effect of angiotensin converting enzyme inhibition on the rate of progression to clinical proteinuria and the rate of change of albumin excretion rates in patients with insulin-dependent diabetes mellitus and persistent microalbuminuria.

Design and Setting.—Randomized, double-blind, placebo-controlled clinical trial of 2 years' duration at 12 hospital-based diabetes centers.

Patients.—Ninety-two patients with insulin-dependent diabetes mellitus and persistent microalbuminuria but no hypertension.

Intervention.—The patients were randomly allocated in blocks of two to receive either captopril, 50 mg, or placebo twice per day.

Measurements.—Albumin excretion rate, blood pressure, glycosylated hemoglobin level, and fructosamine level every 3 months; urinary urea nitrogen excretion every 6 months; and glomerular filtration rate every 12 months.

Results.—Twelve patients receiving placebo and four receiving captopril progressed to clinical proteinuria, defined as an albumin excretion rate persistently greater than 200 μg/min and at least a 30% increase from baseline (P=.05). The probability of progression to clinical proteinuria was significantly reduced by captopril therapy (P=.03 by log-rank test). Albumin excretion rate rose from a geometric mean (95% confidence interval) of 52 (39 to 68) to 76 (47 to 122) μg/min in the placebo group but fell from 52 (41 to 65) to 41 (28 to 60) μg/min in the captopril group, a significant difference (P<.01). Mean blood pressure was similar at baseline in the two groups and remained unchanged in the placebo group but fell significantly, by 3 to 7 mm Hg, in the captopril group. Glycosylated hemoglobin levels and glomerular filtration rate remained stable in the two groups.

Conclusions.—Captopril therapy significantly impeded progression to clinical proteinuria and prevented the increase in albumin excretion rate in nonhypertensive patients with insulin-dependent diabetes mellitus and persistent microalbuminuria.

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DIABETIC RENAL disease is the leading cause of end-stage renal failure in the Western world and affects an even greater proportion of diabetic patients of non-European origin. The cost of caring for patients with diabetes in end-stage renal failure in the United States approached $2 billion in 1991,[1] making the assessment of intervention strategies that may delay or even prevent the development of diabetic nephropathy a priority.

The subset of patients with insulin-dependent diabetes mellitus (IDDM) at risk for nephropathy can be identified by the detection of microalbuminuria. Several longitudinal studies have shown that microalbuminuria reliably predicts overt renal disease as well as cardiovascular disease.[2-6] The progression of microalbuminuria is closely associated with changes in blood pressure (BP) and is affected by the degree of blood glucose control.[7] Antihypertensive therapy reduces the albumin excretion rate (AER) and retards the rate of decline of the glomerular filtration rate (GFR) in patients with IDDM who have overt nephropathy.[8-12] Data from an animal model of diabetes suggest that angiotensin converting enzyme (ACE) inhibition may directly reduce intraglomerular pressure, prevent the increase in the AER, and lessen the development of glomerular histological lesions.[13,14] In patients with IDDM who have microalbuminuria, ACE inhibition has been found to reduce the AER and postpone overt nephropathy. However, the studies reported to date have been either small or unblinded and, with one exception, short-term.[15-17]

We investigated in a randomized, double-blind, placebo-controlled, multicenter trial, the effect of 2 years of captopril therapy on progression to overt nephropathy in patients with IDDM and persistent microalbuminuria.

METHODS

Patients and Study Design

Twelve hospital-based specialist diabetes centers in one Asian and seven European countries participated in the study. Procedures and measurements were specified in a manual of operations.

Patients had to have developed IDDM before the age of 39 years, be between 18 and 55 years of age, and have a duration of diabetes of 4 to 28 years. The AER had to be of 20 to 200 μg/min at screening and in at least two of three consecutive sterile urine samples collected overnight. Arterial pressure had to be lower than 160/95 mm Hg in patients 35 years of age or older and lower than 145/90 mm Hg in patients less than 35 years of age who were taking no antihypertensive medication. Patients receiving or who had been

From the Unit for Metabolic Medicine, United Medical and Dental Schools-Guy's Hospital, London, England (Dr Viberti); Kommunehospitalet University Hospital, Aarhus, Denmark (Dr Mogensen); Helsinki (Finland) University Central Hospital, (Dr Groop); and Bristol-Myers Squibb Co, Princeton, NJ (Mr Pauls).

Participating centers and investigators in the European Microalbuminuria Captopril Study Group are listed in the acknowledgments.

Reprint requests to Unit for Metabolic Medicine, UMDS-Guy's Hospital, Fourth Floor, Hunt's House, London, United Kingdom, SE1 9RT (Dr Viberti).

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treated with antihypertensive medications, nonsteroidal anti-inflammatory drugs, or aldose-reductase inhibitors were excluded. Brittle diabetes, insulin resistance (defined as the need for more than 120 U/d), history of poor compliance, serum creatinine concentration greater than 150 μmol/L (1.7 mg/dL), raised serum potassium concentration, and other renal, endocrine, cardiac, liver, gastrointestinal, or connective tissue diseases were also reasons for exclusion.

Ninety-four patients qualified for the study. Ninety-two agreed to participate and were randomized in blocks of two at each center to receive captopril, 50 mg, or indistinguishable placebo tablets twice per day following a placebo lead-in period of 4 weeks. The choice of a fixed-dose captopril regimen was made to minimize the likelihood of a BP rise in the treated group. All patients gave written informed consent, and the study was approved by the ethics committee of each participating center. Patients underwent laboratory and clinical evaluations every 3 months for 2 years. Treatment compliance was monitored by counting tablets at each visit. No planned modification was made in the patients' usual insulin or dietary regimen throughout the study. Sodium intake remained constant in all patients, ranging between 90 and 110 mmol/d. At each visit, two timed overnight urine samples were collected for analysis of albumin concentration by immunoturbidimetry[18] and creatinine concentration by the Jaffe reaction; a blood sample was collected concurrently for serum analysis of albumin level with a centrifugal analyzer (Cobas Fara, Roche Diagnostics, Welwyn Garden City, United Kingdom), creatinine level by a reaction rate method, and fructosamine level by fructosamine test plus kits (Roche Diagnostics). All these measurements were performed at a central laboratory at Guy's Hospital, London.

The AER was calculated as albumin concentration times urine flow, and albumin clearance was obtained by dividing the AER by the serum albumin concentration. Glycosylated hemoglobin concentrations were determined by local procedures[19] (high-pressure liquid chromatography at four centers, isoelectrofocusing at three centers, ion-exchange chromatography at three centers, electroendosmosis at one center, and Glycaffin GHb kit [Isolab Inc, Akron, Ohio] at one center) at each visit.

Arterial BP (phase I/V) was measured by a random-zero mercury sphygmomanometer with appropriately sized cuff at least 3 hours after the morning dose of insulin and coded medication. With the patient in the sitting position after 5 minutes of rest, two readings, 60 seconds apart, were taken, read to the nearest 2 mm Hg, and the average was used for calculation. Mean arterial pressure was calculated as diastolic pressure plus one-third pulse pressure. If BP exceeded 160/95 mm Hg in patients 35 years old or older or 145/90 mm Hg in patients aged less than 35 years, the measurement was repeated after 2 weeks. If hypertension was diagnosed, a stepped treatment program, including a β-blocker, a diuretic, and a vasodilator, was instituted to maintain BP within the set targets. A 24-hour urine sample was collected at all centers but one every 6 months for measurement of creatinine and urea nitrogen concentrations with a centrifugal analyzer at the central laboratory. The GFR was measured at 12 centers by local procedures (chromium 51–labeled edetic acid plasma clearance at seven centers[20]; technetium Tc 99–labeled diethylenetriamine pentaacetic acid plasma clearance at three centers,[21] and iothalamate sodium I 131 renal clearance at two centers[22]) at baseline and at 12 and 24 months. Fractional albumin clearance was calculated as albumin clearance divided by the GFR.

Primary outcome variables were progression to clinical proteinuria, defined as an AER greater than 200 μg/min and an increase of at least 30% from baseline at two consecutive visits, and rate of change in AER. Secondary study objectives were progression to arterial hypertension and changes in GFR.

Statistics

For demographics and baseline characteristics, differences between treatments were analyzed by χ^2 test for association or by analysis of variance using a linear model that adjusted for numerical differences between treatments within study centers. For weight and body mass index, the linear model also adjusted for sex. Analysis was performed on an intention-to-treat basis, and the effect of treatment on the change from baseline in AER, GFR, BP, glycosylated hemoglobin level, serum fructosamine level, and urinary urea nitrogen excretion was assessed by analysis of covariance to adjust for baseline differences. The model adjusted for numerical differences between treatments within study centers. The AER was log-transformed before analysis because the distribution was positively skewed. Rate of change in AER per year was analyzed by calculating individual regression slopes of log AER over time and comparing the mean slopes for the two treatments with a mixed-model restricted maximum likelihood approach (PROC MIXED, SAS version 6.07).[23] A standardization procedure was used to adjust for variations among local glycosylated hemoglobin assays. Each center's values were first standardized to a z score determined from its normal (reference) range for glycosylated hemoglobin level as follows: $z = [(\text{Glycosylated Hemoglobin} - M)\times 4]/W$, where M is the mean of the upper and lower limits of the glycosylated hemoglobin level and W is the range between the limits. The z scores were then transformed back to percentage units using M and W for the Danish center, selected as a reference, as follows: Glycosylated Hemoglobin = $0.5z + 5.4$. Fisher's Exact Test was used to compare the numbers of patients who progressed to clinical proteinuria. The times to progression were summarized using the Kaplan-Meier procedure to generate a survival curve for each treatment group, the difference between treatments being analyzed by the log-rank test. All tests were two-tailed, and statistical significance was defined as $P \leq .05$. Results are presented as mean ± SEM unless otherwise indicated. Assuming a rate of progression to clinical proteinuria of 30% in the placebo group, the study sample size provides an 80% power to find a statistically significant difference between treatments at the 5% level if captopril were capable of reducing the rate of progression by at least 83%. This sample size would also detect a difference between the mean annual rates of change in AER of the two treatments if the true rates differed by 30% or more.

RESULTS

Table 1 shows the baseline demographic and clinical features of the two groups of 46 patients randomized to receive captopril or placebo. Sex distribution, age, body mass index, duration of diabetes, insulin dose, and smoking habits were similar in the two groups. All patients were of European origin except for two in the captopril group and three in the placebo group, of Oriental origin. Baseline AER, urinary urea excretion, glycosylated hemoglobin level, serum creatinine level, and arterial pressure were similar in the two groups; but the GFR (measured in 82 patients, 42 in the captopril group and 40 in the placebo group) was slightly lower and the serum fructosamine level was higher in the captopril group (Table 2).

Sixteen patients (eight in each group) were withdrawn from the study for the following reasons: unavailable for follow-up, two in the captopril group and five in the placebo group; adverse effects, one in the captopril group and two in the placebo group; pregnancy, three in the captopril group; prohibited therapy, one in the captopril group; low qualifying AER, one in the placebo group; and ac-

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cidental death, one in the placebo group. The average follow-up for these patients was 7.6 months. Four (two in each group) of these 16 patients did not complete the 3-month visit and could not be considered further. Seventy-six patients (39 in the captopril group and 37 in the placebo group) completed the study. Treatment compliance was satisfactory, with an average of 89% of the study medication being taken.

During the 24-month study, 12 patients in the placebo group and four in the captopril group progressed to clinical proteinuria (Fisher's Exact Test, P=.05); the Kaplan-Meier probability of progression was significantly greater in the placebo group (log-rank test, P=.03) (Fig 1). When data were analyzed on an intent-to-treat basis, AER increased from a baseline geometric mean (95% confidence interval [CI]) of 52 (39 to 68) to 76 (47 to 122) μg/min at 24 months in the placebo group but decreased from 52 (41 to 65) to 41 (28 to 60) μg/min in the captopril group, a significant between-group difference (P<.01) (Fig 2). Urinary AER increased significantly, by an average of 18.3% (95% CI, 3.6% to 35.1%) per year from baseline in the patients receiving placebo (P<.05), while the annual increase in the patients receiving captopril, 2.1% (95% CI, −10.2% to 16%), was not significant. The statistically significant differences in AER and rate of change in AER persisted when the potential confounding influence of imbalance in baseline values of AER, BP, glycosylated hemoglobin level, and GFR was tested by analysis of covariance. At 24 months, the fractional albumin clearance increased approximately by 41% in the patients receiving placebo but decreased by 20% in the patients receiving captopril; the difference was statistically significant (P<.05) (Fig 2). A similar proportion of patients with AER in the upper quarter of the distribution (>155 μg/min) at baseline progressed to clinical proteinuria in the two groups (captopril group, 67% [2/3]; placebo group, 60% [3/5]). The percentage of patients with IDDM who progressed to clinical proteinuria with an AER of 155 μg/min or less at baseline was statistically significantly greater in the placebo group (23% [9/39]) than in the captopril group (5% [2/41]; Fisher's Exact Test, P=.02), suggesting that better success might be achieved by starting captopril treatment in patients with lesser degrees of microalbuminuria.

Arterial pressure was similar in the two groups at baseline and did not change appreciably in the placebo group throughout the study. In the captopril group there was a small but significant fall in mean BP of 3 to 7 mm Hg at several time points (Fig 2). Four patients in the placebo group and two patients in the captopril group required antihypertensive therapy. This difference was not significant and did not affect outcome significantly. Mean (95% CI) systolic and diastolic BP at the end of the study were 126 (120 to 133) and 76 (72 to 82) mm Hg, respectively, in the placebo group and 122 (116 to 129) and 74 (70 to 77) mm Hg, respectively, in the captopril group. Independent of treatment, BP rose by 24 months in the group of patients with IDDM who progressed to clinical proteinuria (mean±SEM BP at baseline vs 24 months: 94.0±2.3 vs 99.3±3.7 mm Hg) but tended to fall slightly in those who did not progress (92.3±1.2 vs 89.7±1.66 mm Hg); the difference was significant (P=.03).

The GFR was unchanged during the study in the captopril group; the GFR tended to decrease, although not significantly, in the placebo group and did not differ between the placebo and captopril groups at 12 and 24 months (Fig 2). There was no significant fall in the GFR in patients who, independent of treatment, progressed to clinical proteinuria; their GFR at 24 months was similar to that of patients who did not progress. Serum creatinine (not shown) and glycosylated hemoglobin concentrations remained stable throughout the study in both groups. Serum fructosamine concentration was lower at baseline in the placebo group but increased by 6 months, remaining stable and similar to that of the captopril group thereafter (Fig 3). Urinary urea nitrogen excretion was similar at baseline in both groups (captopril group, 264±16 mmol/d [7.4±0.4 g/24 h]; placebo group, 267±15 mmol/d [7.5±0.4 g/24 h]) and tended to decrease in both groups during the study (P<.01 for the placebo group and P<.05 for the captopril group at 24 months), but no specific treatment effect was found (Fig 3). Serum albumin concentrations were comparable in the two groups both at baseline and during follow-up (data not shown). No significant correlations were found between change in AER and change in BP, glycosylated hemoglobin level, urinary urea nitrogen excretion, or GFR either at baseline or during follow-up in either group. No significant correlation was found between changes in AER and changes in BP within the group of patients with IDDM who progressed to clinical proteinuria. One patient in the captopril group at 3 months had a sudden, unpredictably large increase in AER that persisted at 6 months. A kidney biopsy showed crescentic glomerulonephritis. Persistent cough developed in two patients receiving placebo and a mild skin rash in one patient receiving captopril.

Table 1.—Demographic and Clinical Characteristics of Patients With Insulin-Dependent Diabetes Mellitus and Microalbuminuria

Characteristic	Captopril Group	Placebo Group
No. of patients (M/F)	46 (26/20)	46 (25/21)
Mean age (range), y	32 (19-54)	31 (18-52)
Mean body mass index (range), kg/m²	23.3 (19.3-30.3)	23.9 (18.6-33.0)
Mean duration of diabetes (range), y	16 (5-28)	18 (4-27)
Mean insulin dose (range), U/d	53 (22-96)	52 (28-86)
Smokers, %	56	51

Table 2.—Biochemical and Clinical Characteristics of Patients With Insulin-Dependent Diabetes Mellitus and Microalbuminuria

	Mean (95% Confidence Interval)	
Characteristic	Captopril Group	Placebo Group
Albumin excretion rate, μg/min*	51 (42-61)	59 (47-74)
Glomerular filtration rate, mL/min per 1.73 m²	124† (116-132)	136 (127-145)
Glycosylated hemoglobin, %	9.0 (8.6-9.4)	8.6 (8.1-9.0)
Serum fructosamine, mmol/L	436‡ (407-465)	395 (375-415)
Serum creatinine		
μmol/L	65 (61-68)	64 (61-67)
mg/dL	0.74 (0.69-0.77)	0.72 (0.69-0.76)
Urinary urea nitrogen		
mmol/d	267 (236-298)	264 (231-297)
g/24 h	7.5 (6.6-8.3)	7.4 (6.5-8.3)
Systolic blood pressure, mm Hg	124 (121-128)	124 (120-128)
Diastolic blood pressure, mm Hg	77 (75-79)	76 (74-79)
Mean arterial blood pressure, mm Hg	93 (91-95)	92 (90-95)

*Data are geometric means for the albumin excretion rate.
†Significantly different from placebo (P<.04).
‡Significantly different from placebo (P<.02).

COMMENT

This randomized, double-blind, placebo-controlled, multicenter study dem-

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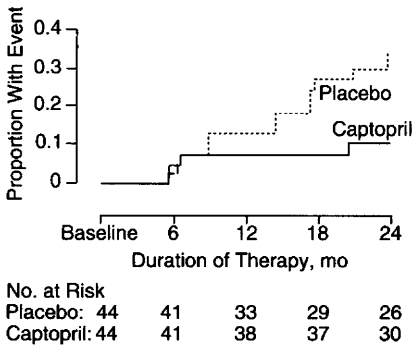

Fig 1.—Probability of progression to clinical proteinuria after treatment with captopril (solid line) or placebo (broken line) in patients with insulin-dependent diabetes mellitus who had microalbuminuria. The difference between the two treatments was statistically significant (P=.03 by log-rank test).

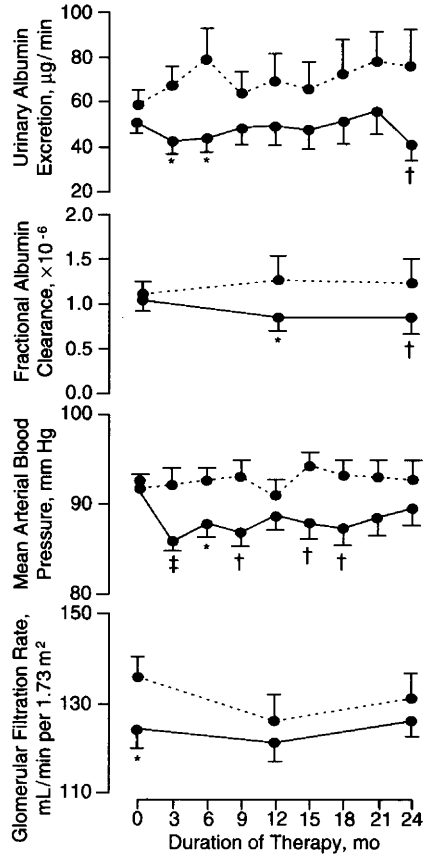

Fig 2.—Albumin excretion rate, fractional albumin clearance, mean arterial pressure, and glomerular filtration rate in patients with insulin-dependent diabetes mellitus who had microalbuminuria and who received either captopril, 50 mg (solid line) or indistinguishable placebo tablets (broken line) twice per day. Values are mean±SEM. Asterisk indicates $P \leq .05$, and dagger, $P \leq .01$, for captopril vs placebo for change from baseline.

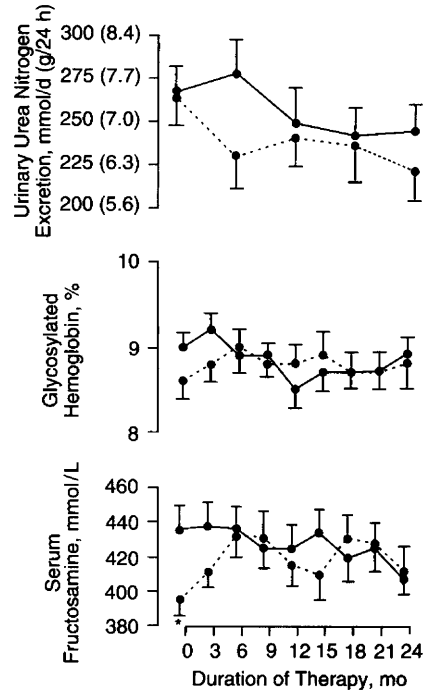

Fig 3.—Urinary urea nitrogen, glycosylated hemoglobin, and serum fructosamine levels in patients with insulin-dependent diabetes mellitus who had microalbuminuria and who received either captopril, 50 mg (solid line) or indistinguishable placebo tablets (broken line) twice per day. Values are mean±SEM. Asterisk indicates $P \leq .05$ for captopril vs placebo at baseline. The urinary urea nitrogen level was lower than baseline by 24 months in both groups, but there were no differences between groups (see text).

onstrates that treatment with the ACE inhibitor, captopril, significantly reduces the rate of progression to clinical proteinuria, the hallmark of overt diabetic nephropathy, in patients with IDDM who have persistent microalbuminuria but who do not have arterial hypertension by the World Health Organization definition. Persistent clinical albuminuria is the harbinger of progressive renal disease in diabetes. Changes in albuminuria are significantly related to the rate of decline in glomerular function in patients with IDDM who have clinical proteinuria,[24] and an increase in proteinuria is associated with deterioration of renal function.[17] Albuminuria, in its own right, is thought to contribute to further glomerular sclerotic damage,[25] and persistent clinical albuminuria has come to be accepted as a surrogate end point in clinical trials aimed at prevention of progression of diabetic kidney disease.[26] Angiotensin converting enzyme inhibition also prevented the significant increase in AER of about 18% per year observed in the placebo group and significantly lowered the fractional clearance of albumin. Although concentrations of plasma renin and angiotensin II were not measured, tablet counts indicated good compliance. Our results with larger numbers of patients confirm the report of an open randomized 4-year trial by Mathiesen et al,[17] who found that captopril therapy prevented the development of persistent proteinuria and lowered the AER in normotensive patients with diabetes who had microalbuminuria. Our findings extend and corroborate other smaller, shorter-term reports of the antialbuminuric effect of ACE inhibitors[15,16,27,28] in patients with IDDM who have microalbuminuria. A stabilizing effect of ACE inhibition on proteinuria and serum creatinine concentration has also been reported in a 5-year controlled clinical trial in normotensive patients with non–insulin-dependent diabetes mellitus who had microalbuminuria.[29]

The reduction in urinary AER is unlikely to have been caused by a decrease in GFR or by changes in serum albumin concentrations, since these two variables did not change appreciably throughout the study. Other variables that could have affected outcome[30,31] are also unlikely to have been responsible for the captopril effect because the degree of glycemic control and protein intake (assessed by urinary urea nitrogen excretion) did not differ in the two groups.

Several mechanisms may have contributed to the antialbuminuric effect of captopril. Although the effect on arterial pressure was numerically small and changes in BP and changes in AER were not correlated in the whole group, an influence of systemic BP changes cannot be excluded. Blood pressure rose independent of treatment category in the group of patients who progressed to clinical proteinuria, and several studies of patients with diabetes who had either microalbuminuria or persistent proteinuria have reported that lowering the systemic arterial pressure using a variety of antihypertensive drugs was associated with reduction in albuminuria.[8,9,15,32,33] In addition, ACE inhibition has been found to reduce efferent arteriolar resistance and to lower the intraglomerular pressure in diabetic rats, preventing the development of albuminuria and restricting glomerulosclerotic lesions.[13,14] This mechanism may also operate in humans. Changes in the selectivity of the glomerular barrier may also result from ACE inhibition, and in diabetic rats given an ACE inhibitor, retardation of glomerular basement membrane thickening has been reported.[34] In patients with diabetes who had moderate to severe proteinuria, ACE inhibition significantly reduced the fractional clearance of high-molecular-weight neutral dextrans, suggesting enhancement

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of the size-selective properties of the glomerular barrier.[35] This effect appears to be independent of systemic BP changes.[36] Finally, experimental studies in other models of chronic renal disease have suggested that ACE inhibition, independent of changes in intraglomerular pressure, may attenuate the growth and accumulation of macromolecules in the mesangium and thus limit traffic of proteins across the glomerular barrier.[37-39]

The GFR did not change significantly in either group and did not differ between groups. These results indicate that an effect on the rate of decline of GFR is unlikely to be detectable before the onset of persistent proteinuria, as also reported by others.[17] Longer observation periods are needed to establish whether ACE inhibition spares the GFR in patients with IDDM who have microalbuminuria. If, as suggested,[25] proteinuria substantially contributes to the progression of renal lesions, a prolonged antiproteinuric effect, such as that seen in the captopril group, might not only prevent or retard the onset of clinical proteinuria but also delay the onset of decline of the GFR.

In conclusion, although additional data on preservation of glomerular function and improved survival are required, our results support the notion that ACE inhibition by captopril therapy retards the progression of renal disease in patients with IDDM who have microalbuminuria.

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European Microalbuminuria Captopril Study Group

Beilinson Medical Centre, Petah Tikva, Israel.—G. Boner, MD, and D. J. van Dyk, MD; *Hospital Germans Trias i Pujol, Badalona, Spain.*—A. Lucas, MD, R. Romero, MD, I. Salinas, MD, and A. Sanmarti, MD; *Helsinki University Central Hospital, Finland.*—A. C. Blomqvist, MD, A. Ekstrand, MD, V. L. Kirsi, MD, V. A. Koivisto, MD, Leif C. Groop, MD, and P. H. Groop, MD; *Hospital Universitario de Granada, Spain.*—F. Escobar, MD, F. E. Jimenez, MD, M. M. Campos-Pastor, MD, M. Muñoz, MD, and M. Gomez, MD; *Istituto Scientifico S. Raffaele, Milan, Italy.*—R. Mangili, MD, G. Pozza, MD, and D. Spotti, MD. *Kommunehospitalet University Hospital, Aarhus, Denmark.*—K. Wurgler Hansen, MD, J. Sandahl Christiansen, MD, F. Klein, and C. E. Mogensen, MD. *Maria Ziekenhuis, Tilburg, the Netherlands.*—L. G. van Doorn, MD; *Medisch Spectrum Twente, Enschede, the Netherlands.*—P. F. M. J. Spooren, MD; *Northwick Park Hospital, Harrow, United Kingdom.*—J. K. Cruickshank, MD; *Rikshopitalet, Oslo, Norway.*—J. Jervell, MD; *Sentralsykehuset i Akershus, Nord Byhagen, Norway.*—P. N. Paus, MD; *United Medical and Dental Schools–Guy's Hospital, London, United Kingdom (coordinating center and central laboratory).*—A. Collins, BSc, G. Viberti, FRCP, and G. Williams, FRCP (external safety monitor); *Watford General Hospital, Watford, United Kingdom.*—G. A. Nelstrop, MD.

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